

Algorithms 2

Vertex-connectivity in graphs

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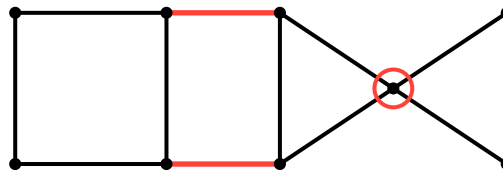
1. Connectivity

- For a graph G , write

$$C(G) := \text{number of connected components of } G$$

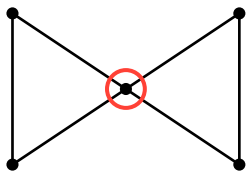
- Let $X \subseteq V(G) \cap E(G)$.

- If $C(G - X) > C(G)$, then X is called a *disconnector*



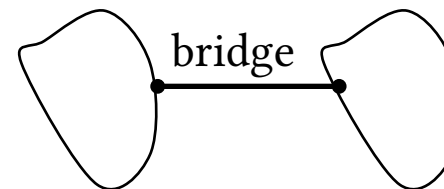
1.1 Graph disconnectors

if $X \subseteq V(G)$, then X is a *vx-disconnector*.



- vx-disconnectors of size 1 are called *cut-vxs*.
- The ends of bridges are cut-vxs.
- The opposite is **not** true!

if $X \subseteq E(G)$, then X is a *edge-disconnector*.



- edge-disconnectors of size 1 are called *bridges*.

Observation 1.1.3 (H.W.)

$e \in E(G)$ is a bridge $\Leftrightarrow e$ does not lie on a cycle of G

2. Whitney's Theorem

Theorem 2.1.1 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

G has no cut-vxs $\Leftrightarrow$ any 2 vertices of G lies on a common cycle.

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proof $\Leftarrow$.

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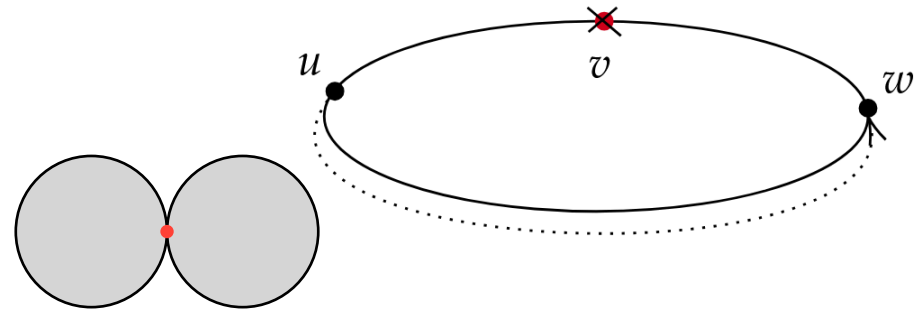
Theorem 2.1.1 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

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proof $\Leftarrow$.

- Assume G has a cut vertex, denoted by v .
 - v separates the graph into 2 components C_1, C_2 .
 - Take $w \in C_1, u \in C_2$ both should hold true
1. A cycle $C := wx_1 \dots x_\ell uw$ exists in G .
 2. If we remove v then u and v are disconnected.



contridiction: If $v \in C$ then removing v or any vertex in C wont separate w from u , and if $v \notin C$ then the path $wx_1 \dots x_\ell u$ still exists.

Theorem 2.1.2 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

G has no cut-vxs $\Leftrightarrow$ any 2 vertices of G lies on a common cycle.

$\Rightarrow$

- Suppose G has no cut-vxs.
- We prove by induction on

$\delta_{G(u,v)} :=$ length of the shortest path $u \rightsquigarrow v$ in G

- Basis:
 - Let $u, v \in V(G)$ have $\delta_G(u, v) = 1$
 - If $uv \in E(G)$ is a bridge, the u, v are cut-vxs. **contradiction.**
 - From the observation uv lies on a cycle.

Observation 2.1.3 (H.W.)

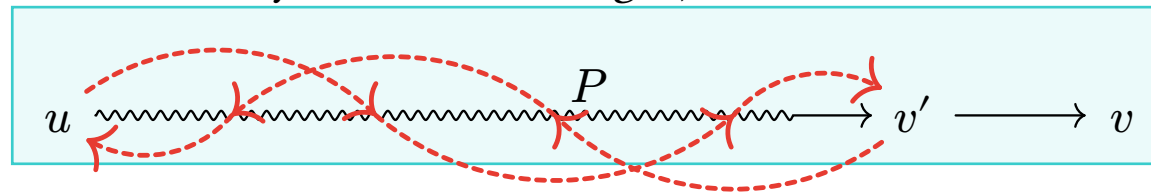
$e \in E(G)$ is a bridge $\Leftrightarrow$ e does not lie on a cycle of G

Theorem 2.1.4 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

G has no cut-vertices $\Leftrightarrow$ any 2 vertices of G lies on a common cycle.

- Suppose G has no cut-vertices.
- Suppose that the claim hold true for all pairs of vertices u, v such that $\delta_{G(u,v)} < k$
- Consider a pair $u, v \in V(G)$ satisfying $\delta_G(u, v) = k$
- Let P be the shortest path $u \rightsquigarrow v$, and let v' denote the vertex before v on P .
- As $\delta_G(u, v') = k - 1$, there is a cycle C containing u, v'



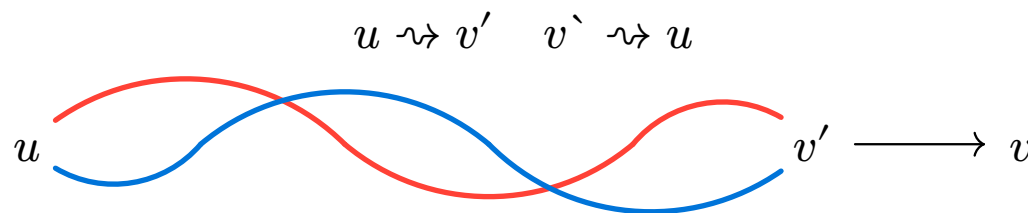
- If $v \in V(C)$ we are done

Theorem 2.1.5 (Whitney's Theorem)

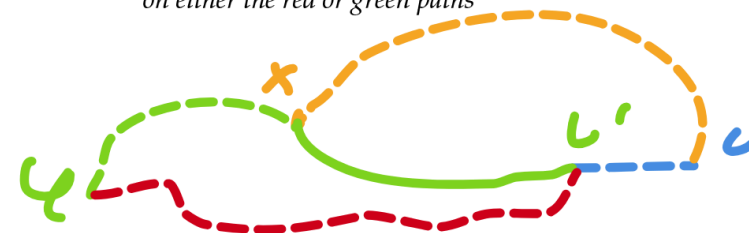
- G - a connected graph with $v(G) \geq 3$

G has no cut-vertices $\Leftrightarrow$ any 2 vertices of G lies on a common cycle.

- Otherwise, we can divide C into 2 arcs



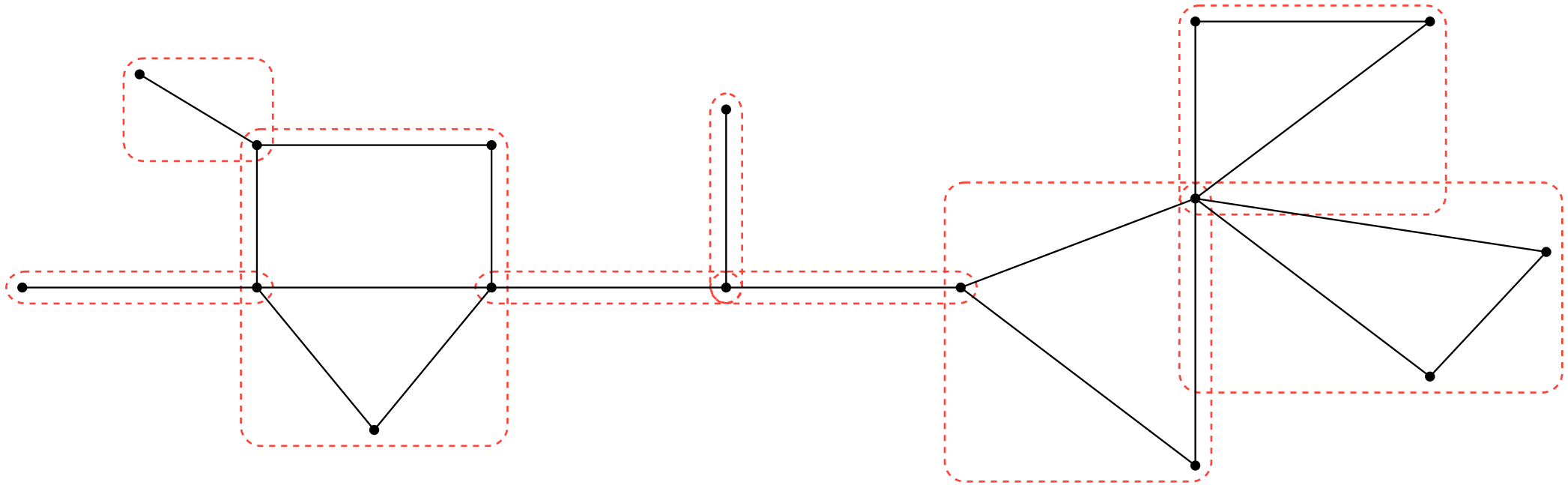
x is the first vertex v meets
on either the red or green paths



- As G has no cut-vertices, $G - v'$ is still connected,
- There is a path from v to a vertex x that lies on one of the arcs)
- W.L.O.G, let $x \in u \rightsquigarrow v'$ **take the shortest path from v to any of the arcs!**
- Then the cycle $u \rightsquigarrow x \rightsquigarrow v \rightarrow v' \rightsquigarrow u$ is a cycle, and contains both u and v .

3. Consequences of Whitney's Theorem

Definition 3.1.1 A maximal connected subgraph of G containing no cut-vertex is called a block of G



Theorem 3.2.1 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

G has no cut-vxs $\Leftrightarrow$ any 2 vertices of G lies on a common cycle.

- Any two blocks of G meet in at most one vertex.
- If two block do meet, their intersection vertex is a cut-vertex.
- The blocks of G partition $E(G)$ (H.W.).
- Every cycle of G lies in precisely one block of G (H.W.).

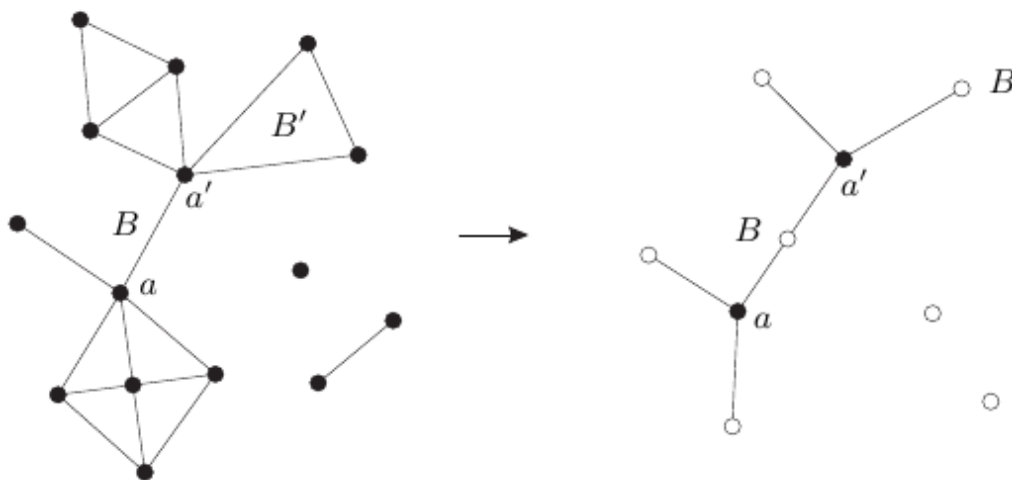
Given a graph G define the following auxiliary graph:

- $B(G) := \{B : B \subseteq G \text{ is a block of } G\}$
- $C(G) := \{v : v \in V(G) \text{ is a cut-vertex of } G\}$

Let $BC(G)$ denote the following graph

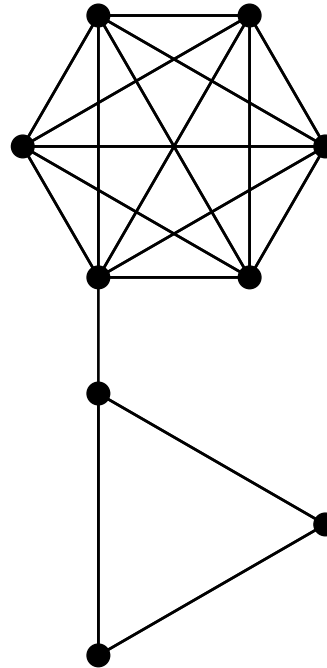
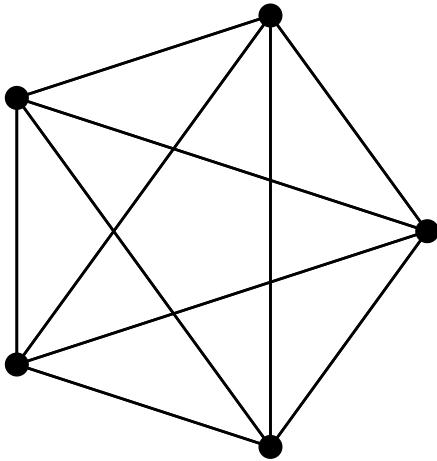
- The vertices of $BC(G)$ are $B(G) \cap C(G)$
- The edges of $BC(G)$ are $\{\{B, v\} : B \in B(G), v \in C(G) \text{ and } v \in B\}$

Theorem 3.3.1 if G is connected then $BC(G)$ is a tree



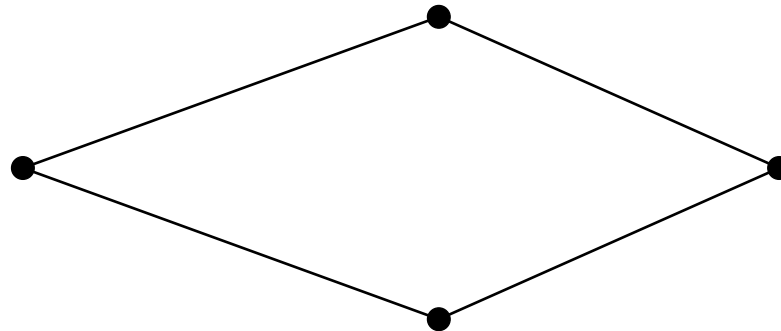
4. *Assesing connectivity*

Question 4.1.1 Given these graphs, which one is more connected?



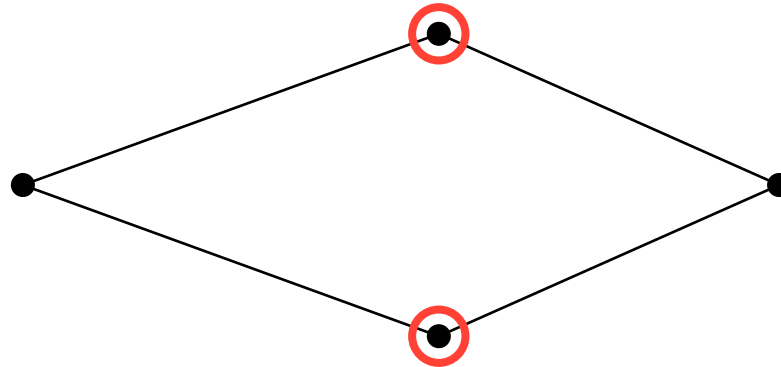
Definition 4.2.1

- $0 \leq k \in \mathbb{N}$
- G - graph with $v(G) > k$
- If $G - X$ is connected $\forall X \subseteq V(G), |X| \leq k - 1$ then G is called *k -connected*



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- To be k -connected, $v(G) > k$ must hold
 - What about K_1 ?
 - We assume it is connected
 - But our definition K_1 is **not** connected!
 - We'll fix this later

Definition 4.3.2 The largest $k \in \mathbb{Z}_{\geq 0}$ for which G is k -connected is denoted by $\kappa(G)$

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- $\kappa(G) \geq 1$
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- Cycles with 3 or more vertices are **2-connected**

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- $\kappa(K_r) = r - 1$

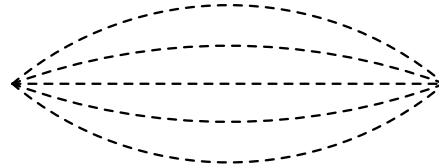
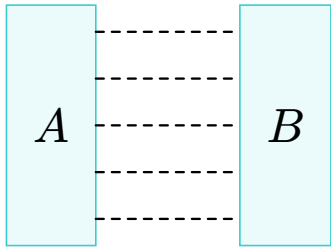
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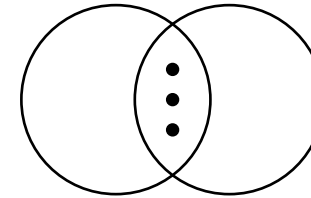
Again $\kappa(K_1) = 0$, is annoying, can we fix it?

Definition 4.4.1

- G a graph
- $A, B \subseteq V(G)$
- let $\rho_G(A, B) :=$ the number of vx-disjoint (A, B) -paths



Here $|A| = |B| = 1$



$A \cap B \neq \emptyset$ also can happen

 Remark

Vertices in A and B counts toward the vertices of the paths.

If $|A| = 1$ or $|B| = 1$ then we count the number of paths from the neighbours of A or B respectively.

Definition 4.5.1 (Path connectivity) G is *k -connected* if $\rho_G(u, v) \geq k \quad \forall \{u, v\} \in \binom{V(G)}{2}$

Definition 4.5.2 (Vertex connectivity) Graph G with $v(G) > k > 1$ is *k -connected* if for every $X \subseteq V(G)$ with $|X| \leq k - 1$ the graph $G - X$ is connected. If $v(G) = 1$ then G is *1-connected*.

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So whats better: vertex connectivity or path connectivity ?

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So whats better: vertex connectivity or path connectivity ?

Menger's theorem tells us that they are the same!

Definition 4.6.1 The minimum size of a vx set X such that $G - X$ is disconnected or has a single vx is called the *vx-connectivity* of G and is denoted by $\kappa(G)$

Definition 4.6.2 A graph G is called *k-connected* if $\kappa(G) \geq k$

 Remark

Other than the complete graph:

A graph is k -connected $\Leftrightarrow$ all of its vx-cuts are of size $\geq k$

The complete graph has no vx-cuts of any size, this is where the 2nd part of the definition comes in

5. Menger's theorem

Definition 5.1.1

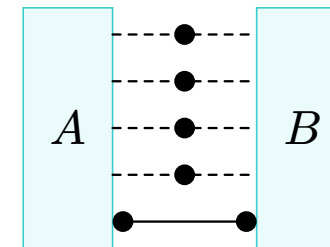
- $A, B \subseteq V(G)$
- Denote by $\kappa_G(A, B)$ the size of the **least** (A,B)-vx-disconnetor

Theorem 5.1.2 (Menger's theorem) Let G be a graph and let $A, B \subseteq V(G)$ non empty. Then

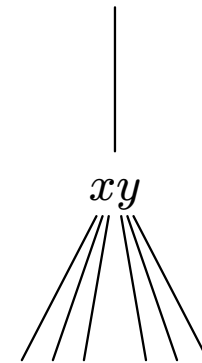
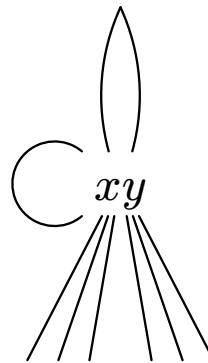
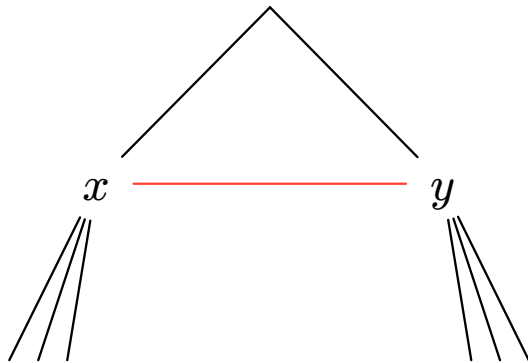
$$\kappa_G(A, B) = \rho_G(A, B)$$

- You know this by the name “min-cut-max-flow”
- $\kappa_G(A, B) \geq \rho_G(A, B)$ is trivial
- We need to prove

$$\kappa_G(A, B) \leq \rho_G(A, B)$$



- Consider the edge $e = xy$
- To contract e :
 - remove e
 - merge x and y into one vertex v_{xy}
 - connect v_{xy} to $N_G(x) \cap N_G(y)$, remove any duplicant edges that may arise



- Denote by G/e the graph resulting from contracting e

Theorem 5.3.1 (Menger's theorem) Let G be a graph and let $A, B \subseteq V(G)$ non empty. Then

$$\underbrace{\kappa_G(A, B) = \rho_G(A, B)}_{\text{The goal is to show } \leq}$$

- Set $\kappa := \kappa_G(A, B)$ and $\rho := \rho_G(A, B)$
- Proof by induction on $e(G)$
- If $e(G) = 0$ the graph is disconnected
 - $\rho = |A \cap B| = \kappa$

Goal : Show that $\kappa_G(A, B) \leq \rho_G(A, B)$.

I.H: For every graph G' with $e(G') < e(G)$ and every $A', B' \subseteq V(G')$ non empty, we have $\kappa_{G'}(A', B') \leq \rho_{G'}(A', B')$

- Assume $e(G) > 0$, let $e = xy \in E(G)$ be an arbitrary edge
- G/e satisfies the I.H. but A and B are not defined consistently in $G \setminus e$. **(What if x or y is in A or B ?)**.

Define $A_e \subseteq V(G/e)$ as follows

- If $x, y \notin A$, then $A_e := A$
- if $x \in A, y \notin A$, then $A_e := (A \setminus \{x\}) \cup \{V_e\}$
- if $x \notin A, y \in A$, then $A_e := (A \setminus \{x\}) \cup \{V_e\}$
- if $x, y \in A$, then $A_e := (A \setminus \{x, y\}) \cup \{V_e\}$
- Define B_e in the same manner.

5.3 Menger's theorem

Case 1: Suppose that G/e has κ v_x -disjoint (A_e, B_e) -paths

- **Goal:** Show that every such path exists in G .
 - Any path not containing v_e exists in G .
 - suppose that there is a path in the linkage containing v_e .

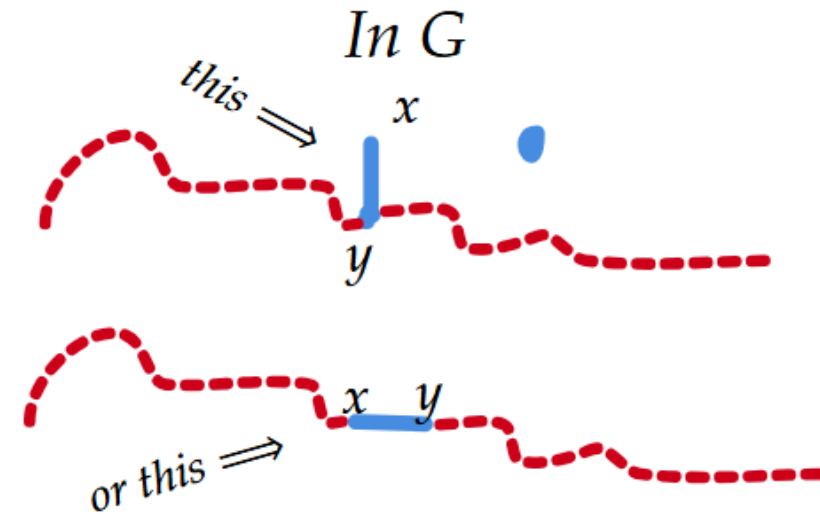
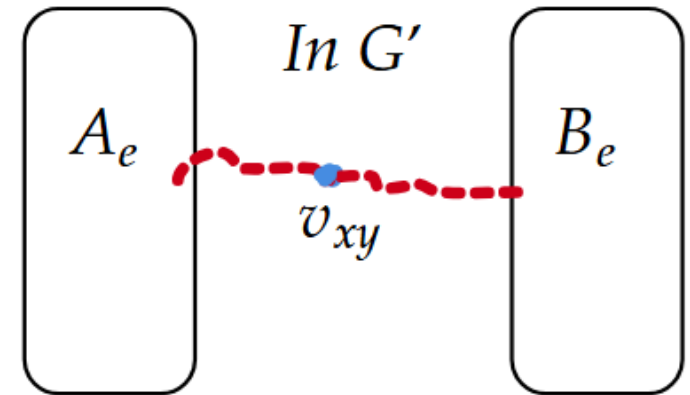
xy can be expanded in the path

----- $\in N_G(x)$ ----- x ----- y ----- $\in N_G(y)$ -----
 ----- $\in N_G(y)$ ----- y ----- x ----- $\in N_G(x)$ -----

xy cannot be expanded in the path

----- $\in N_G(y)$ ----- x ----- y ----- $\in N_G(y)$ -----
 ----- $\in N_G(x)$ ----- x ----- y ----- $\in N_G(x)$ -----

- In the second case, only one of the vertices lie on the original path.

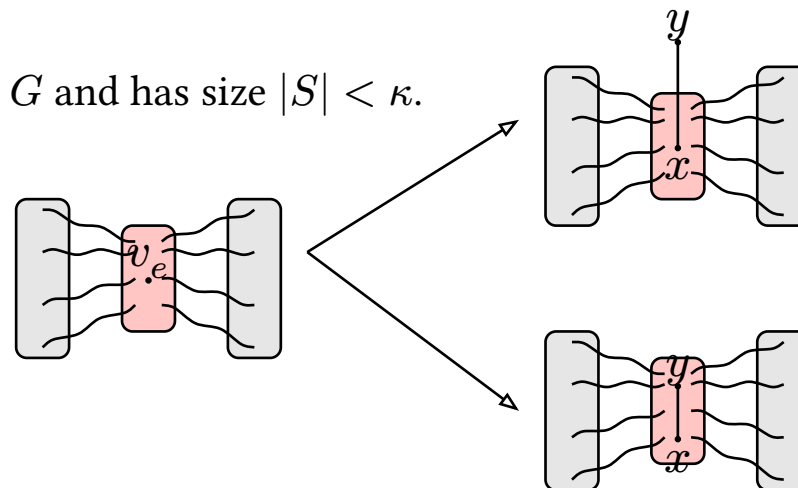


Claim 5.4.1 If S is vx-disconnector of (A_e, B_e) in G/e then a vx-disconnector of (A, B) in G exists where

$$S_G \subseteq (S \setminus v_e) \cup \{x, y\}.$$

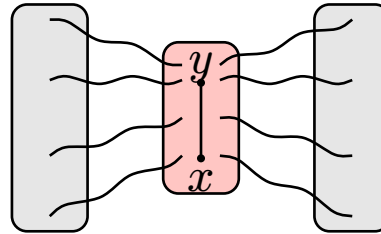
Case 2: $\rho_{G \setminus e}(A_e, B_e) < \kappa$

- By the I.H $\kappa_{G \setminus e}(A_e, B_e) < \kappa$.
- Let S be a vx-disconnector of (A_e, B_e) .
- $v_e \in S$ must hold:
 - If $v_e \notin S$, then S is also a vx-disconnector of (A, B) in G and has size $|S| < \kappa$.
- Any (A_e, B_e) -vx-disconnector in G/e is of size $\kappa - 1$.
 - Otherwise $|S_G| \leq |S| + 1 < \kappa - 1 + 1 = \kappa$.



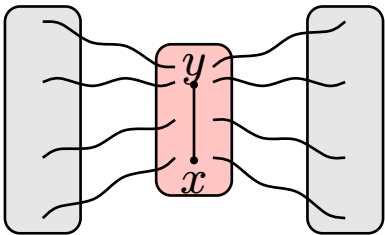
We are now in this position:

- The edge xy lies inside the minimum vx -disconnecter of (A, B) in G .

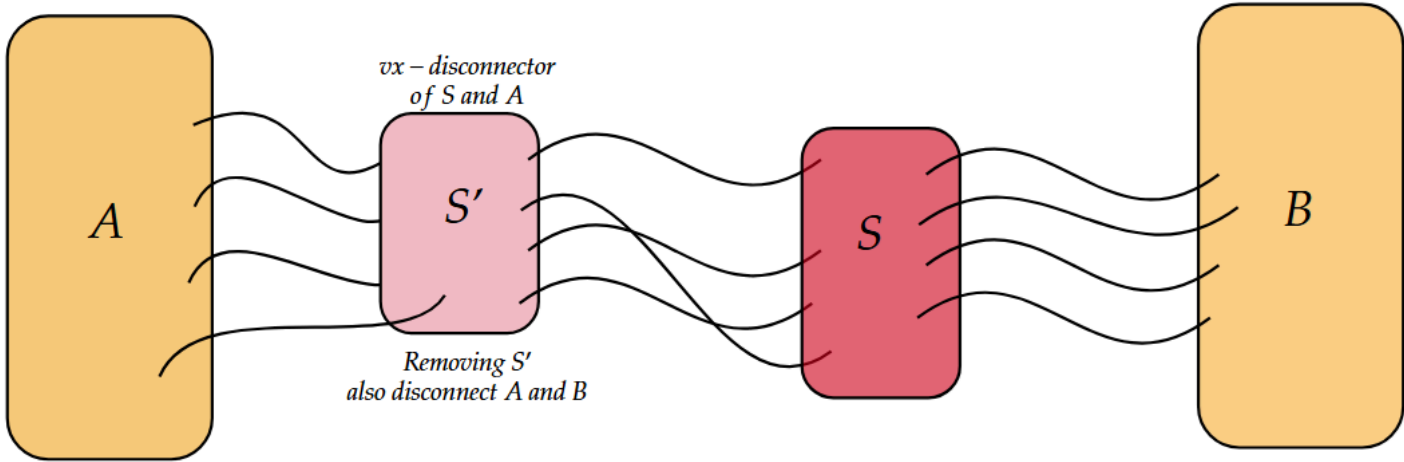


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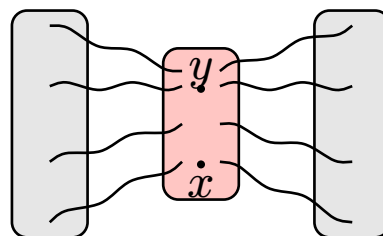


Observation: The size of the least vx -disconnecter of (A, S) and (S, B) is at least κ .

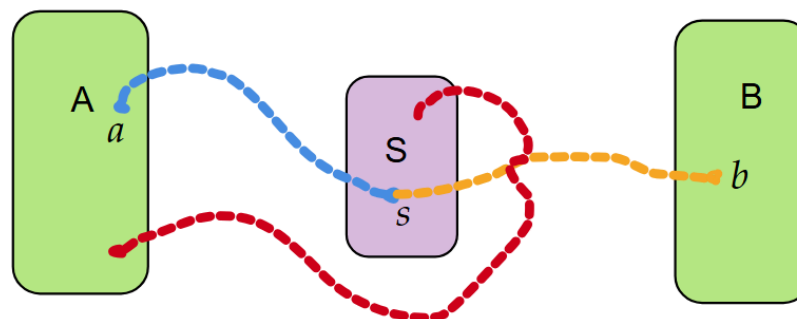


- Look at $G - e$: **(removing the edge e rather than contracting it!)**
- By the induction hypothesis.

$$\rho_{G-e}(A, S_G) = \kappa_{G-e}(A, S_G) \geq \kappa \quad \text{and} \quad \rho_{G-e}(S_G, B) = \kappa_{G-e}(S_G, B) \geq \kappa$$



- There must exist κ vx-disjoint (A, S_G) -paths and κ vx-disjoint (S_G, B) -paths in $G - e$.
- For any path $a \rightsquigarrow s$ and $s \rightsquigarrow b$, where $a \in A, s \in S, b \in B$.
 - The path $a \rightsquigarrow s \rightsquigarrow b$ is vx-disjoint (A, B) -path in G .



Red path cannot occur in G ,
As then S doesn't disconnect A and B .

6. Implication of Menger's theorem

Corollary 6.1.0.1 (H.W.)

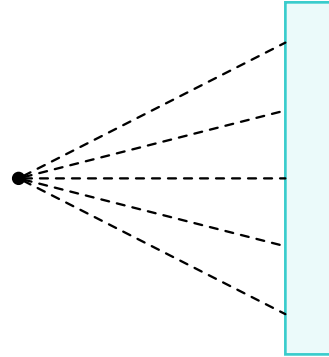
- Let G be a graph. Then

$$\kappa(G) = \min\{\rho_G(u, v) : u, v \in V(G), u \neq v\}$$

Lemma 6.1.1 (The Extension Lemma H.W.)

- G - k -connected graph
- H obtained from G by:
 - Adding a new vertex
 - Connecting the new vertex to k arbitrary vertices
- Then H is k -connected,

- Given $x \in V(G)$ and $B \subseteq V(G)$ s.t. $x \notin B$, we write (x, B) -fan to denote:



Lemma 6.1.2 (Fan Lemma H.W)

- G - k -connected graph
- $x \in V(G)$ and $B \subseteq V(G) \setminus \{x\}$

Then G has an (x, B) -fan if size $\min\{k, |B|\}$

Theorem 6.2.1 (Dirac's Theorem)

- $2 \leq k \in \mathbb{N}$
- G - k -connected graph
- $S \subseteq V(G)$ such that $2 \leq |S| \leq k$

Then G contains a cycle constaining S

- By induction on k .
- for $k = 2$, the claim holds by whitney's theorem.

Theorem 6.2.2 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

G has no cut-vxs $\Leftrightarrow$ any 2 verticies of G lies on a common cycle.

I.H: If G is a $k - 1$ connected graph then for every $S \subseteq V(G)$ with $|S| \leq k - 1$ there is a cycle containing S .

- Assume $k \geq 3$
 - Let $x \in S$ be arbitrary and set $T := S \setminus \{x\}$.
- $|T| \leq k - 1$.
- G is k -connected so $G - x$ is at least $k - 1$ connected.
 - By the I.H. there is a cycle C containing T .
- let F be an (x, C) -fan in G of size $\min\{k, v(C)\}$.
- T partitions C into $k - 1$ arcs.
- At least two of the paths in the fan land in the same arc on C .

Theorem 6.2.1 (Dirac's Theorem)

- $2 \leq k \in \mathbb{N}$
- G - k -connected graph
- $S \subseteq V(G)$ such that $2 \leq |S| \leq k$

Then G contains a cycle constaining S

